

EXP 6.1

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COURSE CODE: ECA0910

COURSE NAME: DIGITAL SIGNAL PROCESSING

PRACTICAL LAB RESULTS

CODE:

```
clc;
clear;
close all;
%% Parameters
Fs = 1000;      % Sampling frequency
f = 50;         % Signal frequency
T = 0.1;        % Duration
A = 1;          % Amplitude
t = 0:1/Fs:T;
x = A*sin(2*pi*f*t);
Levels = [8 16 32];
figure(1)
subplot(4,1,1)
plot(t,x,'b','LineWidth',1.5)
title('Original Sinusoidal Signal')
xlabel('Time (s)')
ylabel('Amplitude')
axis([0 0.1 -1.1 1.1])
grid on
for k=1:length(L Levels)
    L = Levels(k);
    % Step size
    Delta = 2/(L-1);
    % Uniform Quantization
    xq = round((x+1)/Delta)*Delta - 1;
    % Saturation
    xq(xq>1)=1;
    xq(xq<-1)=-1;
    XQ{k}=xq;
    subplot(4,1,k+1)
    stairs(t,xq,'b','LineWidth',1.5)
    title(['Quantized Signal ( ' num2str(L) ' Levels)'])
    xlabel('Time (s)')
    ylabel('Amplitude')
```

```

axis([0 0.1 -1.1 1.1])
grid on
end
figure(2)
for k=1:length(Levels)
    e=x-XQ{k};
    subplot(3,1,k)
    plot(t,e,'LineWidth',0.8)
    title(['Quantization Error (' num2str(Levels(k)) ' Levels)'])
    xlabel('Time (s)')
    ylabel('Error')
    xlim([0 0.1])
    grid on
end

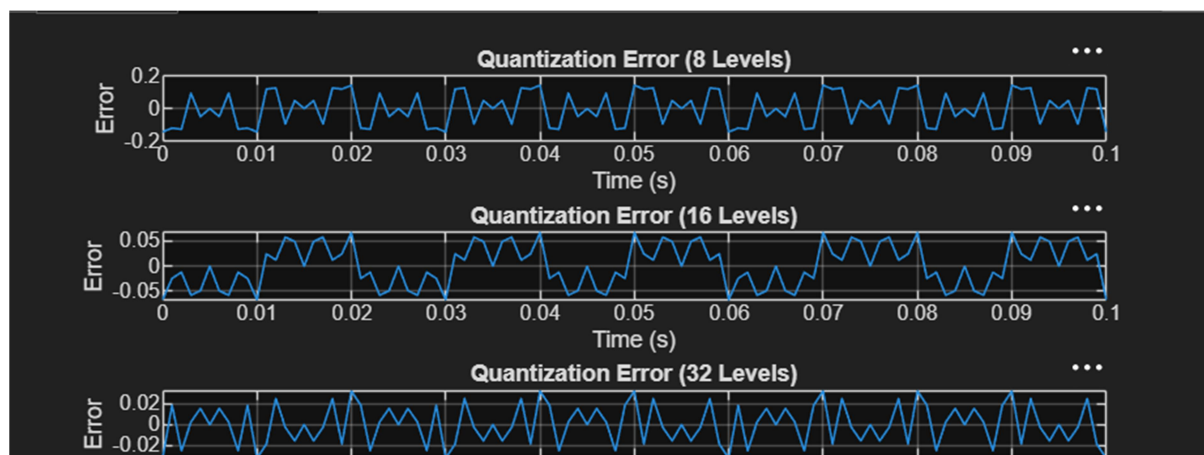
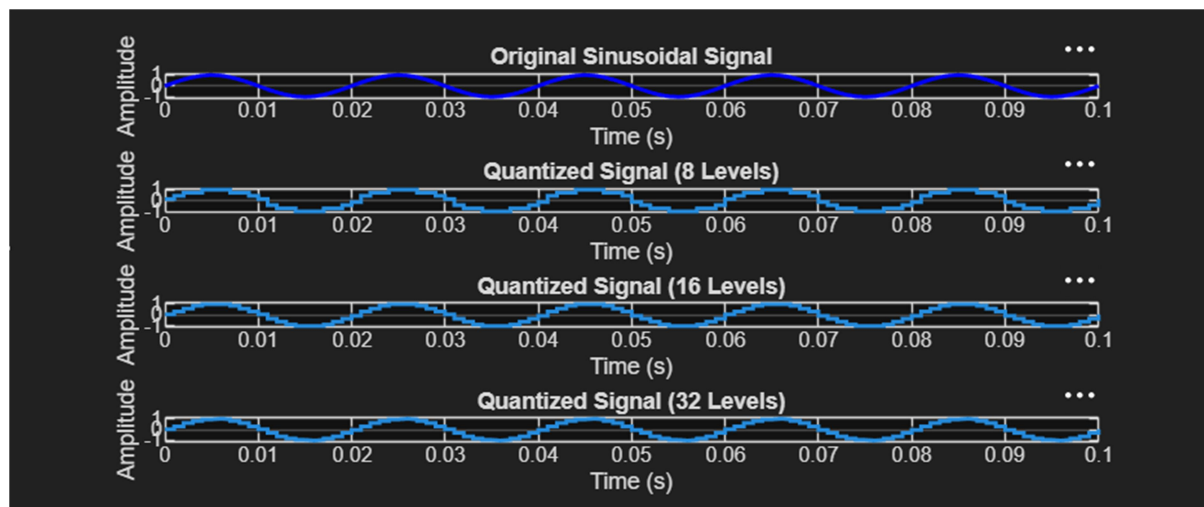
```

OUTPUT:

Sampling Frequency, $f_s = 1000\text{Hz}$

Signal Frequency, $f = 50\text{Hz}$

Time Duration = 0.1 s



EXP 6.2

CODE:

```
clc; clear; close all
n=0:79;
x=sin(2*pi*n/40);
q=8;
xt=floor(x*q)/q;
xr=round(x*q)/q;
et=x-xt;
er=x-xr;
% Fig 1
figure
plot(n,x,'b',n,xr,'r','LineWidth',1.5)
grid on
title('Original vs Quantized')
legend('Original','Quantized')
% Fig 2
figure
plot(n,x,'b',n,xt,'r',n,xr,'g','LineWidth',1.5)
grid on
title('Truncation vs Rounding')
legend('Original','Truncated','Rounded')
% Fig 3
figure
stem(n,et,'r')
hold on
stem(n,er,'g')
grid on
title('Quantization Error')
legend('Truncation','Rounding')
% Fig 4
figure
bar([mean(et.^2) mean(er.^2)])
set(gca,'XTickLabel',{'Truncation','Rounding'})
grid on
title('MSE Comparison')
```

```
ylabel('MSE')
```

